Aditya Kumar

7707062214 |2512.adityasingh@gmail.com | LinkedIn | GitHub | Website | Seattle, WA

Summary

Software Engineer specializing in scalable cloud applications, distributed systems, and microservices. Experienced in API design, CI/CD automation, and deploying fault-tolerant systems on AWS, Azure, and GCP. Skilled in container orchestration, performance optimization, and integrating AI/ML for intelligent automation.

Education

Master's in Computer Science: Georgia Institute of Technology, Atlanta, GA **GPA: 3.9/4**
Bachelor of Engineering in Information Science: CMR Institute of Technology, Bangalore, India **GPA: 8.76/10**

Technical Skills

Languages & Frameworks: Java, Python, C#, C/C++, Scala, SQL, Spring Boot, React.js, HTML, CSS, REST APIs
Cloud & DevOps: AWS , Azure ML, GCP, Kubernetes, Docker, Ansible, Terraform, Jenkins, GitHub Actions
Data & Streaming: Kafka, Redis, PostgreSQL, OpenSearch, Airflow, MLflow, Kubeflow, Druid
Architecture & Patterns: Microservices, Event-Driven Architecture, RAG Pipelines, CI/CD, A/B Testing
ML Integration: AI Agents, LLMs, NLP, Predictive Modeling, Time Series Forecasting, Recommendation Systems

Experience

Allstate Insurance | Lead Software Engineer Oct 2020 - Aug 2024

Earned two promotions in three years by consistently demonstrating leadership and innovation.

- **Architected a distributed HR AI platform** using AWS ECS microservices with FastAPI, API Gateway, and Cognito; **sustained 99.9% uptime** and **sub-300 ms latency** at **300+** concurrent users.
- Developed a **microservices-driven** storage forecasting platform on Kubeflow, **reducing model deployment time by 40%** and enabling more accurate, scalable infrastructure planning.
- Built a **React.js + Ansible self-service provisioning portal** backed by REST APIs, enabling automated service orchestration across 20+ cross-functional teams.
- Designed a **distributed real-time anomaly detection system** using Airflow for orchestration, Kafka for event streaming, and ECS for service hosting; cut false positives by 70% and **reduced detection latency to 30 s**.
- Developed **service-oriented time series monitoring** components for mainframe CPU clusters, deployed across multiple availability zones to ensure resiliency, **reducing incident detection time by 50%**.
- Automated network and storage provisioning workflows via **API-first design, scaling to 2,000+ monthly requests** with **98%** faster turnaround.
- Built **distributed Kubeflow pipelines for parallel ML training**, integrating with service-oriented architectures and achieving **3 times faster training cycles**.
- Developed **modular containerized Python services for incident classification**, cutting **detection and classification time by 40%** across a distributed architecture.

Projects

Disease Diagnosis Pipeline with Multimodal Data Fusion | GitHub Link

- Engineered a **modular preprocessing pipeline** to convert echocardiography DICOM videos into tensors and integrate with cleaned tabular data, ensuring separation of concerns and **scalable batch inference**.
- Automated environment setup with bootstrap scripts and symlink management; built a trainer abstraction to standardize training/evaluation workflows, improving developer experience and maintainability.

Real-Time Cognitive Load Monitoring System | Github Link

- Implemented a **real-time streaming pipeline** to ingest physiological signals via OSC, process inputs with pretrained transformers, and trigger alerts when elevated cognitive load persisted.
- Structured **lifecycle management** for encoders, scalers, and models into reusable modules, enabling reproducible deployments and simplified model updates.

Research Experience

Graduate Research Assistant, Georgia Institute of Technology Nov 2024 – Apr 2025

- Developed **scalable implementations** of Conjunctive Normal Form (CNF) for neural networks, enabling analysis of larger models and improving system efficiency.
- Optimized network architectures and collaborated with team members to enhance **model reliability and security.**